

INTERNATIONAL SOCIETY FOR ROCK MECHANICS
COMMISSION ON
STANDARDIZATION OF LABORATORY AND FIELD TESTS



SUGGESTED METHODS FOR DETERMINING
THE STRENGTH OF ROCK MATERIALS IN TRIAXIAL COMPRESSION

INTRODUCTION

The Commission on Standardization of Laboratory and Field Tests on Rock was appointed in 1967. Subsequent to its first meeting in Madrid in October 1968, the Commission circulated a questionnaire to all the members of the International Society for Rock Mechanics, the answers received clearly showing a general desire for standardized testing procedures. At a further meeting in Oslo in September 1969, tests were categorized and a priority for their standardization was agreed upon, as given in Table 1.

It was also decided that research tests, including many of the rock physics tests, were beyond the scope of standardization. Subsequent meetings were held in Belgrade in September 1970, in Nancy in October 1971, in Lucerne in September 1972, in Katowice in October 1973, in Denver in September 1974, in Minneapolis in September 1975 and in Salzburg in October 1976. At the Lucerne meeting the Commission was subdivided into two committees, one on standardization of laboratory tests and the second on the standardization of field tests.

The present document has been produced by the Committee on Standardization of Laboratory Tests. The present document covers Category II (1) in Table 1.

It should be emphasized that the purpose of these "Suggested Methods" is to specify rock testing procedures and to achieve some degree of standardization without inhibiting the development or improvement of techniques.

Any person interested in these recommendations and wishing to suggest additions or modifications should address his remarks to the Secretary General, International Society for Rock Mechanics, Laboratório Nacional de Engenharia Civil, Avenida do Brasil, Lisboa, Portugal.

Acknowledgements—The following persons contributed in the drafting of these "Suggested Methods": U. W. Vogler (South Africa), K. Kovari (Switzerland).

TABLE 1. TEST CATEGORIES FOR STANDARDIZATION

Category I: Classification and Characterization

Rock material (laboratory tests)

- (1) Density, water content, porosity, absorption.*
- (2) Strength and deformability in uniaxial compression; point load strength.*
- (3) Anisotropy indices.
- (4) Hardness, abrasiveness.*
- (5) Permeability.
- (6) Swelling and slake-durability.*
- (7) Sound velocity.*
- (8) Micro-petrographic descriptions.*

Rock mass (field observations)

- (9) Joint systems: orientation, spacing, openness, roughness, geometry, filling and alteration.*
- (10) Core recovery, rock quality designation and fracture spacing.
- (11) Seismic tests for mapping and as a rock quality index.
- (12) Geophysical logging of boreholes.*

Category II: Engineering Design Tests

Laboratory

- (1) Determination of strength envelope (triaxial and uniaxial compression and tensile tests).*
- (2) Direct shear tests.*
- (3) Time-dependent and plastic properties.

In situ

- (4) Deformability tests.*
 - (5) Direct shear tests.*
 - (6) Field permeability, ground-water pressure and flow monitoring; water sampling.
 - (7) Rock stress determination.*
 - (8) Monitoring of rock movements, support pressures, anchor loads, rock noise and vibrations.
 - (9) Uniaxial, biaxial and triaxial compressive strength.
 - (10) Rock anchor testing.*
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* Asterisks indicate that final drafts on these tests have been prepared.

Suggested Methods for Determining the Strength of Rock Materials in Triaxial Compression

1. SCOPE

This test is intended to measure the strength of cylindrical rock specimens subjected to triaxial compression. This provides the values necessary to determine the strength envelope and from this the value of the internal friction angle ϕ and the "apparent" cohesion C may be calculated.*

2. APPARATUS

The apparatus consists essentially of three parts (See Fig. 1): a triaxial cell, a loading device and a device for generating confining pressure.

2.1. A triaxial cell

This comprises:

(a) Test specimens shall be right circular cylinders be placed in order to apply the confining pressure. The body of the cell should have an air bleeder hole and a connection for a hydraulic line.

(b) A flexible jacket of suitable material to prevent the hydraulic fluid from entering the specimen, and which shall not significantly extrude into abrupt surface pores.

(c) The triaxial cell shall be filled with hydraulic fluid, than C30 shall be placed at both specimen ends. The diameter of the platens shall be between D and $D + 2$ mm, where D is the diameter of the specimen.

* No provision has been made for drainage of the pore water, nor for the measurement of its pressure. In certain rock types (e.g. shales) and under certain conditions the pore water pressure may influence the results. In such cases it is advisable to conduct tests on specimens with different degrees of saturation, e.g. saturated, oven dried at 105°C or any other. A comparison of the results allows an estimation of the influence of the pore water.

Such conditions shall be reported in accordance with "Suggested method for determination of the water content of a rock sample", Method 1, ISRM Committee on Laboratory Tests, Document 2, Final Draft, November 1972.

† The concave halves of the spherical seats in triaxial machines usually have no freedom of movement in the direction perpendicular to the specimen axis. In order to align itself, the specimen must have two spherical seats. This is contrary to the uniaxial compression test where the top concave seat half has freedom in the lateral direction and where only one spherical seat is thus required.

‡ The procedure for, and time intervals between verifications are usually given in the National Standard specifications, e.g. ASTM E4; DIN 51300 and B.S. 1610.

§ In order to fulfil the requirements of section 2.3. (b) for the accuracy of the pressure indicating device, it may be necessary to use two or more interchangeable pressure indicating devices having different ranges. Their accuracy will generally have to be 4–5 times better than that of the pressure to be maintained.

The thickness of the platens shall be at least 15 mm or $D/3$. Surfaces of platens should be ground and their flatness should be 0.005 mm.

(d) Spherical seats which are incorporated in each of the platens.† The curvature centre of the seat surfaces should coincide with the centre of the specimen ends.

2.2. A loading device for applying and controlling axial load

(a) A suitable machine shall be used for applying, controlling and measuring the axial load on the rock specimen. It shall be of sufficient capacity and capable of applying the load at a rate conforming to the requirements as set out in section 4(e). It shall be verified at certain time intervals‡ and shall comply with the accepted national requirements such as prescribed in either ASTM Methods E 4, Verification of Testing Machines; British Standard 1610, 1964, Grade A or Deutsche Normen DIN 51 220 and DIN 51 223, Klasse 1 and DIN 51 300.

(b) The spherical seat of the loading machine, if any, and if it is not complying with specification 2.1 (d) above, shall be removed or placed in a locked position, the two loading faces of the machine being parallel to each other.

2.3. Equipment for generating and measuring the confining pressure

This includes:

(a) A hydraulic pump or pressure intensifier or other system of sufficient capacity and capable of maintaining constant confining pressure within 2% of the desired value.

(b) A pressure indicating device§ (pressure gauges or pressure transducers) which shall be accurate enough to allow the above to be observed or recorded.

3. PREPARATION OF THE TEST SPECIMEN

(a) Test specimens shall be right circular cylinders having a height to diameter ratio of 2.0–3.0 and a diameter preferably of not less than NX core size (approximately 54 mm). The diameter of the specimen should be related to the size of the largest grain in the rock by the ratio of at least 10:1.

(b) The ends of the specimen shall be flat to 0.02 mm and shall not depart from perpendicularity to the longi-

tudinal axis of the specimen by more than 0.001 radian (about 3.5 minutes) or 0.05 mm in 50 mm.

(c) The sides of the specimen shall be smooth and free of abrupt irregularities and straight to within 0.3 mm over the full length of the specimen.

(d) The use of capping materials or end surface treatments other than machining is not permitted.

(e) The diameter of the test specimen shall be measured to the nearest 0.1 mm by averaging two diameters measured at right angles to each other at about the upper-height, the mid-height and the lower-height of the specimen. The average diameter shall be used for calculating the cross-sectional area. The height of the specimen shall be determined to the nearest 1.0 mm.

(f) Samples shall be stored for no longer than 30 days, in such a way as to preserve the natural water content, as far as possible, until the time of specimen preparation. Following their preparation, the specimens shall be stored prior to testing for 5–6 days in an environment of $20^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and $50\% \pm 5\%$ humidity.* This moisture condition shall be reported in accordance with "Suggested method for determination of the water content of a rock sample", Method 1, ISRM Committee on Laboratory Tests, Document No. 2, Final Draft, November 1972.

(a) The number of specimens to be tested as well as the number of different confining pressure values should be determined from practical considerations, but at least five specimens per rock sample are preferred in addition to the uniaxial compressive strength tests conducted according to the relevant specifications.†

4. PROCEDURE

(a) The cell shall be assembled with the specimen aligned between steel platens and surrounded by the jacket.‡ The specimen, the platens and the spherical seats shall be accurately aligned so that each is coaxial with the others.

(b) The spherical seats should be lightly lubricated with mineral oil.

(c) The triaxial cell shall be filled with hydraulic fluid, allowing the air to escape through an air bleeder hole. Close air bleeder hole.

(d) The cell shall be placed into the axial loading device (Fig. 1).

(e) The axial load and the confining pressure must be increased simultaneously and in such a way that axial stress and confining pressure be approximately equal, until the predetermined test level for the confining pressure is reached. Subsequently, the confining pressure shall be maintained to within 2% of the prescribed value.

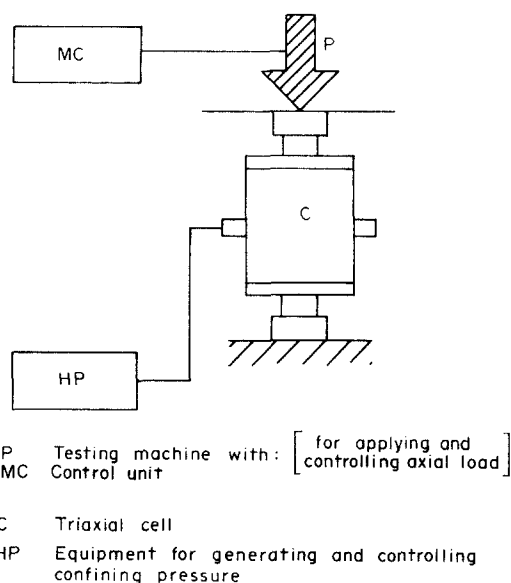


Fig. 1. Block diagram showing test arrangement for determining the triaxial compressive strength.

(f) The axial load on the specimen shall then be increased continuously at a constant stress rate such that failure will occur within 5–15 min of loading. Alternatively the stress rate shall be within the limits of 0.5 to 1.0 MPa/s.

(g) The maximum axial load and the corresponding confining pressure on the specimen shall be recorded.

5. CALCULATIONS

(a) The compressive strength of the specimen shall be calculated by dividing the maximum axial load, applied to the specimen during the test, by the original cross-sectional area.

(b) The confining pressures and the corresponding strength values for the different specimens are plotted with the confining pressures as abscissae and strengths as ordinates (Fig. 2).

(c) A strength envelope is obtained by fitting a mean curve to the above points. From practical considerations it may be advisable to fit a straight line to only the most relevant part of the curve, or to fit several straight lines to different parts of the curve. Each straight line is characterized by calculating its gradient (tangent of the inclination) m and its Y intercept, b . In each case the range in which the respective straight line is valid must be shown on the abscissa.

(d) Using parameters m and b , the internal friction angle ϕ and a value for the "apparent" cohesion C (in the sense of Coulomb's failure theory) may be calculated using the formulae:

$$\phi = \arcsin \frac{m-1}{m+1}; \quad C = b \frac{1 - \sin \phi}{2 \cos \phi}.$$

6. REPORTING OF RESULTS

The report should include the following:

(a) Lithologic description of the rock.

* See footnote * on p. 48.

† The test programme, i.e. the choice of the confining pressure values, depends on practical considerations regarding the purpose of the tests.

‡ The acceptable triaxial cells in use differ widely. No exact guidelines as how to install the rock specimen or how to assemble the cell can therefore be given.

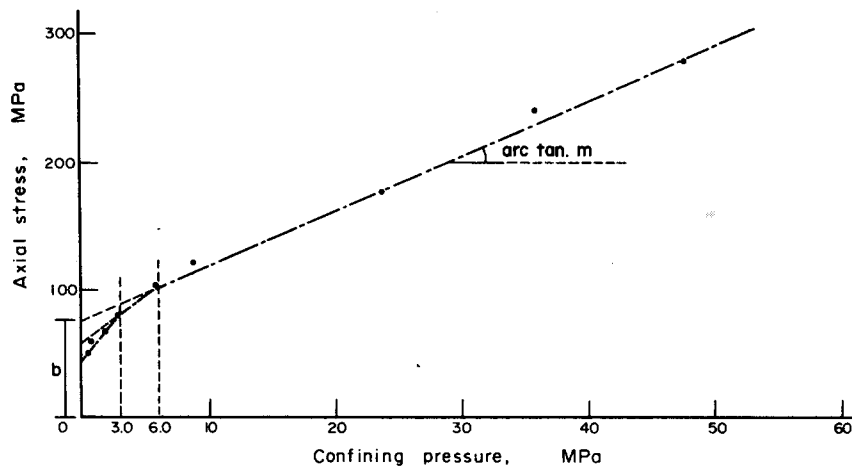


Fig. 2. Strength envelope.

(b) Orientation of the axis of loading of specimen with respect to anisotropy, bedding planes, foliation, etc.

(c) Source of sample including: geographic location, depth and orientation, data and method of sampling and storage history and environment.

(d) Water content and degree of saturation at time of test.

(e) Test duration and/or stress rate.

(f) Date of testing and description of testing machine, triaxial cell and equipment for creating and measuring of confining pressure.

(g) Number of specimens tested.

(h) Any other observations, e.g. mode of failure or available physical data, e.g. specific gravity, porosity, etc., citing the method of determination of each.

(i) A table giving specimen number, specimen height, specimen diameter, confining pressure and the corresponding axial strength to 3 significant figures.

(j) The plot of axial strength vs confining pressure as discussed in section 5(b) (Fig. 2).

(k) A table giving the values of C and ϕ together with the range of confining pressures in which they are valid.

(l) Should it be necessary in some instances to test specimens that do not comply with the above specifications, these facts shall be noted in the test report.

REFERENCES

1. International Society for Rock Mechanics. Committee on Laboratory Tests. Suggested method for determining the uniaxial compressive strength of rock material. Document No. 1, first revision (March 1977).
2. ASTM. Standard method of test for triaxial compressive strength of undrained rock core specimens without pore pressure measurements. ASTM Designation D 2664-67.
3. Kovari K. & Tisa A. *Höchstfestigkeit und Restfestigkeit von Gesteinen im Triaxialversuch*. Institut für Strassen- und Untertagebau an der ETH Zürich. Mitteilung Nr. 26.